

The Trading Strategy — explained simply, in depth

What the strategy does and why, in plain words and worked examples

This is the full guide to the trading idea behind `Backtesting/engine.py`. It goes step by step with real numbers, but keeps the language simple. If you can picture a stretched rubber band snapping back, you can follow all of it.

The whole journey in one line: **work out what's normal** → **spot when a price gap is stretched too far** → **check the snap-back is worth more than the trading cost** → **fade it** → **ride the reversion through fair to a ~1.5-wiggle overshoot** → **get out at the right moment**.

1. The big idea (in one minute)

We trade **spreads** — the *gap* between two related oil prices. For example, the price of oil for July delivery minus the price for August delivery. That gap might be **\$0.74**.

We do **not** bet on whether oil goes up or down. We only bet on whether that **gap returns to normal** after it gets stretched.

Picture the gap as a rubber band. Most of the time it rests at a comfortable length. Now and then a burst of buying or selling **stretches** it — too wide or too narrow. History shows it usually **snaps back**. So when we see a stretched gap, we bet on the snap-back. That's the whole strategy:

When the gap is unusually stretched, bet that it returns to normal.

Everything else is about doing that carefully — measuring "stretched" honestly, and getting in and out at sensible points.

Why bet on the gap, not on oil itself? Because we buy one month and sell another at the same time, so the big swings in the *overall* oil price cancel out. What's left is a small, steady, well-behaved relationship — far easier to predict than "will oil go up?"

2. What we actually trade

We **screen seven** gaps, all built from **WTI** (American oil) and **Brent** (global oil) futures, in three shapes:

- **Calendar gaps** — the same oil, two different months (e.g. July vs August WTI). A pure bet on whether near-term oil is getting tighter or looser than later oil.
- **Butterflies** — a three-month combination that checks whether the *middle* month is out of line with its neighbours. The most surgical, market-neutral of the three.
- **Brent vs WTI** — the gap between the two kinds of oil, driven by shipping and regional supply differences.

But we **trade only the three** that show a *persistent, after-cost* edge once we prune the rest: the **Brent–WTI arb**, the **WTI butterfly (M1–M2–M3)**, and the **WTI M2–M3** calendar. The others — notably the Brent *front* calendars — looked fine gross but didn't keep enough net of cost to earn a seat, so we leave them on the screen and off the book.

In every case we **buy one contract and sell another at the same time**, so we own no outright oil — only the gap. One contract is 1,000 barrels.

3. The key question — what is "fair value," and can we borrow Phase 2's model?

To know whether a gap is stretched, we need a reference: its **fair value** (its "normal" level). The project already built a clever fair-value model in **Phase 2** — a statistical model that predicts each spread from real-world causes (oil stockpiles, the US dollar, market fear, the season). The obvious question: **can we just reuse that here?**

We looked carefully, and the answer is **no** — for three solid reasons:

1. **Its ingredients aren't in this data.** The Phase 2 model needs fundamentals — inventory levels, refinery activity, the dollar, the fear gauge. The intraday feed we're trading contains **only oil prices** — none of those ingredients. So the model simply **can't be computed** here.
2. **Its number is out of date.** Phase 2's fair value for the WTI July–August gap is about **2.18**. That same gap is trading at about **0.74** today — the contracts have re-priced since. Forcing the old number on would scream "this gap is cheap by 1.4!" *forever* — a permanent false signal, not a real one.
3. **It barely moves intraday.** Those fundamentals update daily (or weekly), so over a few days the model is basically a flat line. It can't explain the minute-by-minute wiggles we're actually trading, which come from order flow, not from slow fundamentals.

So we estimate fair value from the gap's own recent behaviour — see the next section. This isn't a step *down* from Phase 2; it's the *right tool for the timeframe*. Phase 2 still contributes: its proven track record for each gap (how often that gap historically snapped back) is carried in as a **confidence score** on every trade. We just don't use its *price* as the reference, because it can't be one here.

4. Step one — work out what "normal" looks like

We look at the gap's recent history — the **last 24 readings, about six hours** of trading — and take their **average**. That average is our "normal" (the fair value) for right now. As new readings arrive, the average rolls forward, so "normal" keeps up with a slowly drifting gap.

We also measure how much the gap **normally wiggles** — its typical jiggle from reading to reading. Call one of those a "**wiggle**." A gap that normally jumps around a lot needs to move *a lot* to count as stretched; a calm gap needs only a little. Measuring in wiggles puts every gap — calm or jumpy — on the **same fair ruler**.

5. Step two — measure how stretched it is

We turn "how far from normal" into one number, counted in wiggles. Traders call it a **z-score**; you can read it as "*how many normal wiggles away from normal are we?*"

- Around **0** = sitting right at normal.
- **+2 or more** = unusually **high** — the gap is "**rich**" (too wide).
- **-2 or less** = unusually **low** — the gap is "**cheap**" (too narrow).

Worked example. Normal is **0.74**, one wiggle is **0.05**, and right now the gap is **0.84**.

Distance from normal = **0.10**. In wiggles: $0.10 \div 0.05 = 2.0$. So the stretch reading is **+2.0** — stretched wide, right at our trigger.

6. Step three — the entry rule

Two conditions, both of which must hold: **fade the gap once it is stretched to 2 wiggles or more — but only if the expected dollar move back to fair clears about twice the realistic trading cost.**

- If it's **rich** (stretched wide, +2 or more) → **sell** the gap, betting it narrows back.

- If it's **cheap** (stretched narrow, -2 or less) → **buy** the gap, betting it widens back.

We picked **2 wiggles** because it's a genuinely unusual stretch — unusual enough that the snap-back is a real edge, but common enough to give a decent number of trades. (Lower, like 1, trades more but on weaker, noisier signals; higher, like 3, waits for rare extremes.)

The cost gate is the second condition, and it's what makes the strategy actually pay. A 2-wiggle stretch isn't the same dollar size on every gap: on a calm, low-volatility spread one wiggle might be only half a cent, so a full 2σ stretch is worth a cent or two — too small to beat the bid-ask, a guaranteed loser after fees. On a livelier gap the same 2σ is worth real money. So before we fade, we ask: *is the expected move back to fair at least $\sim 2\times$ what it costs to get in and out?* If yes, we trade; if no, we skip it, however stretched it looks. Because the gate is measured in dollars against each gap's own volatility, it's **volatility-adaptive** — it automatically waves through the worthwhile stretches and filters out the cheap ones.

7. Step four — the exit rules

Once we're in, we leave for the **first** of four reasons:

1. **It worked — take profit at the overshoot.** We do **not** bail the moment the gap touches normal. Stretched gaps usually *overshoot* — they sail through fair and out the other side before settling — and that overshoot is worth a large slice of the profit. So instead of closing at fair, we **ride the reversion through fair to a ~ 1.5 -wiggle overshoot on the far side** (we exit at a stretch of -1.5 , i.e. `Z_EXIT = -1.5`) and bank the whole arc.
2. **We were wrong — stop loss.** Instead of reverting, the gap stretched **further against us**, to 3.5 wiggles. We accept we're wrong and cut it.
3. **It's taking too long — time stop.** If a trade hasn't resolved within **about 12 hours**, we stop waiting and close it. This avoids sitting in dead trades that drift nowhere.
4. **The day is ending — flatten.** We never hold through the overnight or weekend gap, and we never carry across a contract-roll break (too much can happen while the market is closed or re-pricing). We close out before the break.

The shape of the bet. We enter at $+2$ wiggles and ride all the way through 0 down to -1.5 — a **~ 3.5 -wiggle** reward — while cutting at $+3.5$, a **1.5-wiggle** risk. So the reward leg is more than twice the risk leg, *and* mean-reversion wins more often than it loses — that combination is the edge. Capturing the overshoot, rather than ducking out at fair, is what turns the gross signal into a comfortably net-profitable one.

8. Step five — how big is each trade?

One unit per trade by default. Simple.

This is a deliberate choice. The point of this backtest is to answer "*does the signal work?*" — so we keep the size fixed and small, and report the **raw per-unit result**. One contract is 1,000 barrels, so a \$1 move in a gap is worth \$1,000.

There's one **optional** lever on top: a **2-unit scaling toggle**. When it's on, we add a *second* unit on the very deepest dislocations — stretches of **2.75 wiggles or more** — on the theory that the biggest stretches snap back hardest. It's **off by default** (the headline numbers are all 1-unit), and we report what it would have added separately so you can see the signal cleanly first and the scaled version second.

An honest note. An earlier version tried clever position-sizing (risking a fixed 1% of a \$250,000 account per trade). It technically worked, but it multiplied every position ~100×, which blew the profit *and* the losses up into scary six-figure numbers that obscured whether the underlying signal was any good. So we stripped it back out. **Sizing is a separate decision** you layer on *after* you trust the signal — not part of testing the signal. Here we test the signal, cleanly, at one unit.

We keep a pretend \$250,000 only as a starting line for the running profit/loss chart.

9. A full trade, start to finish

1. **A gap gets stretched.** WTI July–August normally rests near 0.74. A burst of buying pushes it to 0.84 — a stretch of about **+2.0 wiggles**, rich. ✓ It's past our trigger.
2. **The cost gate clears.** The expected move back to fair is 0.10 per unit (\$100) — comfortably more than twice what it costs to get in and out — so the trade is worth doing. ✓
3. **We enter — SELL the gap at 0.84**, betting it falls back toward 0.74 and beyond.
4. **The confidence score** is set from Phase 2's track record for this gap (WTI calendars reverted ~75% of the time historically) plus how stretched it is — here, a solid ~80/100.
5. **It reverts — and overshoots.** A few hours later the gap doesn't just touch 0.74, it sails *through* it down to about 0.665 — a **-1.5-wiggle** overshoot on the far side. That's our take-profit ($Z_{EXIT} = -1.5$). **We close and pocket the whole arc** (the gap fell 0.175, × \$1,000 per unit per \$1 = a solid win — markedly more than if we'd ducked out at fair).

A **losing** trade differs only at the end: the gap pushes *up* to 3.5 wiggles instead, our stop fires, and we take a small, pre-planned loss. Or it stalls and the 12-hour time stop closes it near break-even.

10. How we keep score

For every trade we record: which gap, buy or sell, the in/out prices and times, why we exited, how long we held, the profit or loss, and the worst/best point it passed through.

From all the trades together, three numbers matter most:

- **Profit factor** — total dollars won ÷ total dollars lost. Above **1.0** = making money; **3.0** = won three dollars for every one lost. The fairest single score, because a "win-often-lose-big" trick can fake a high win rate but **cannot** fake a high profit factor.
 - **Expectancy** — the average profit per trade. The real bottom line.
 - **Biggest dip (drawdown)** — the worst peak-to-valley fall in the running total; the "pain" measure.
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11. The cost check — does it survive trading fees?

By default we run **before costs** (the pure signal). But we can also charge **realistic trading costs** (the small fee on every buy and sell).

The honest finding: **the strategy survives costs comfortably and stays strongly net-profitable.**

Net of a realistic 1¢-per-leg fee it **keeps about 89% of its gross** — only about a tenth of the edge is eaten by execution. That's the pay-off from the three design choices that an earlier, naïve version lacked: the **cost gate** refuses the stretches too small to beat the fee, the **pruning** keeps only the structures that hold up after cost, and the **overshoot exit** banks the far side of each reversion so each winning trade carries a bigger margin over its fixed cost. Put together, they turn what used to look like a break-even, cost-bound idea into one that is genuinely **tradeable** — the prediction works *and* the dollars clear the till.

12. The results

12a. The headline — the historical intraday backtest

The real verdict comes from the long run: **5.4 years of 15-minute WTI/Brent data**, the three traded structures, 1 unit per trade.

Measure	Result	What it means
Trades	5,388	stretched-gap opportunities taken over 5.4 years
Gross profit	+\$2,231,606	total per-unit dollars made (before costs)
Net (after a 1¢/leg fee)	+\$1,979,446	keeps ~89% of gross — costs barely dent it
Profit factor	4.93	won \$4.93 for every \$1 lost
t-statistic	30.8	the edge is real, not luck — overwhelmingly significant
Profitable years	every year 2021–2026	positive in all six calendar years
Max net drawdown	≈ −\$12,750	worst dip along the way, net of cost

With the optional **2-unit scaling toggle ON**, net rises to about **+\$2,622,400**.

Per structure (gross / net @ 1¢ per leg):

Structure	Trades	Gross	Net @ 1¢	Profit factor
Brent–WTI	3,103	\$1,617,504	\$1,493,384	5.50
WTI butterfly (M1–M2–M3)	916	\$362,753	\$289,473	5.19
WTI M2–M3	1,369	\$251,348	\$196,588	3.07

All three carry their own weight after cost. (The Brent *front* calendars that used to drag the book have been pruned out, so there's no longer a money-losing structure to flag.)

12b. The live snapshot — the mentor's company feed

The live company feed is only a **few days** of 15-minute data, so it's a small, noisy slice — useful as a freshness check, not a verdict:

Measure	Result
Trades	~23
Gross profit	~\$1,980
Net @ 1¢ per leg	~\$820
Profit factor	~2.8

These are small, honest, per-unit numbers — the raw signal, not a leveraged book.

12c. The daily cross-check — the historical *daily* backtest

There's a second, independent backtest on **daily** data that uses a completely different fair-value reference: the **Phase 2 fundamental regression** (inventories, the dollar, the fear gauge, the season), run **walk-forward / out-of-sample** so each day is scored only on data that came before it. The rules differ to suit the daily horizon: **enter on a $\geq 1.5\sigma$ residual** (price minus the model's fair value), **exit at 0.5σ** , with a **3σ stop** and a **20-day** time stop. We prune to the same kind of five-structure survivor set — again **dropping the two losing Brent front calendars**.

Measure	Result
Trades	198
Gross profit	\$117,183
Net @ 2¢ per leg	\$94,143 (keeps ~80%)
Profit factor	2.34
Profitable years	all six

The key rule of the two models. The fundamental regression fair value is **daily-only** — its ingredients update once a day, so it can't price the minute-by-minute wiggles. The **intraday** strategy therefore *must* use the rolling, price-derived fair value from §4, never the fundamental one. Two timeframes, two fair values, by design.

13. The honest limits

The headline is now a **5.4-year, 5,388-trade backtest that is net-profitable in every year**, so the old worries that "there isn't much data" and "costs decide it / it's break-even" **no longer apply to the historical results**. They survive only as a caveat on the few-day **live snapshot** (§12b), which really is too short to call. With that reframing, the honest limits are:

1. **The live snapshot is still thin.** A few days of company-feed data is a freshness check, not a verdict — only the long historical backtests carry the weight.
2. **The cost figures are illustrative.** 1¢/leg (intraday) and 2¢/leg (daily) are reasonable stand-ins, but your real fees, financing and spread will differ.
3. **15-minute closes are not real fills.** We trade at bar closes; live execution would face slippage and partial fills the backtest can't see.
4. **"Normal" is a simple average.** A single sharp spike can tug the average a little; a more robust measure is a possible refinement.
5. **Oil only.** The data is WTI and Brent crude — no diesel, petrol, or other products.
6. **One-unit raw signal.** The headline is the raw per-unit signal, not a sized portfolio; sizing and risk limits (and the optional 2-unit toggle) are a separate layer on top.

The robustness battery

Because the intraday numbers are strong enough to invite "too good to be true," we put them through a dedicated stress test — `analytics/robustness.py`, which writes `robustness.json`. It checks the strategy five ways:

- **Per-year P&L** — is it profitable in *every* year, not just on average? (Yes — all of 2021–2026.)
- **t-statistic** — is the average edge statistically real rather than luck? (Yes — $t \approx 30.8$.)
- **Monte-Carlo drawdown** — reshuffling the trade order, how bad could the worst dip plausibly get?
- **Walk-forward structure selection** — if we'd picked the traded structures using only past data at each point, would we still have chosen these three? (Yes.)
- **Parameter-sensitivity grid** — does the edge hold across neighbouring entry/exit/stop settings, or does it live on one lucky knob? (It holds across the grid.)

The intraday strategy **passes all five**, which is what lets us treat the §12a numbers as a real result rather than a curve-fit.

14. Mini-glossary (plain words)

- **Spread / gap** — the price difference between two related contracts, traded as one bet.
 - **Rich / cheap** — a gap that's unusually wide (rich) or narrow (cheap) versus normal.
 - **Normal / fair value** — the gap's recent resting level (here, the average of recent readings).
 - **Wiggle** — the gap's typical jiggle; the unit we measure "stretch" in.
 - **Stretch / z-score** — how many wiggles away from normal we are right now.
 - **Fade** — to bet *against* a move (sell what just jumped up, buy what just dropped).
 - **Snap-back / mean reversion** — the tendency of a stretched gap to return to normal.
 - **Cost gate** — the rule that we only fade when the expected move back to fair clears about twice the trading cost; volatility-adaptive, so it skips stretches too small to be worth it.
 - **Overshoot** — when a reverting gap sails *through* normal and out the far side; we exit at a ~1.5-wiggle overshoot to bank the whole arc, not just the move back to fair.
 - **Stop loss** — the pre-set point where we admit we're wrong and cut the trade.
 - **Take profit** — closing a winning trade at the overshoot, once it has reverted through fair.
 - **Profit factor** — dollars won ÷ dollars lost (above 1 = profitable).
 - **Drawdown** — the worst drop from a high point to a later low; the "pain" measure.
 - **Gross vs net** — results before costs (gross) versus after costs (net).
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15. How to run it

```
python Backtesting/engine.py          # before costs (the pure signal)
python Backtesting/engine.py --slip 0.01 # charge a per-leg fee; reports net too
python Backtesting/engine.py --live     # keep re-running on the latest data
```

Each run writes the full trade-by-trade story to `out/trades_log.md`, a spreadsheet to `out/trades.csv`, and feeds the live dashboard. The plain-English overview of the whole project is in `../PROJECT_REPORT.md`; the engine's methodology is in `IDEATION.md`.